

Supplier databases

```
create database supplier;
```

```
use supplier;
```

```
create table supplier(sid int primary key ,  
sname varchar(20) not null,city varchar(20));
```

```
create table parts( pid int primary key ,pname varchar(20)  
not null,color varchar(10));
```

```
create table catalog(sid int ,pid int ,cost int ,  
foreign key(sid) references supplier(sid),  
foreign key(pid) references parts(pid));
```

```
desc supplier;
```

```
desc parts;
```

```
desc catalog;
```

```
insert into supplier values(10001,'Acme Widget','Bangalore');
```

```
insert into supplier values(10002,'Johns','Kolkata');
```

```
insert into supplier values(10003,'Vimal','Mumbai');
```

```
insert into supplier values(10004,'Reliance','Delhi');
```

```
select * from supplier;
```

```
insert into parts values(20001,'Book','Red');
```

```
insert into parts values(20002,'Pen','Red');
```

```
insert into parts values(20003,'Pencil','Green');  
insert into parts values(20004,'Mobile','Green');  
insert into parts values(20005,'Charger','Black');
```

```
select * from parts;
```

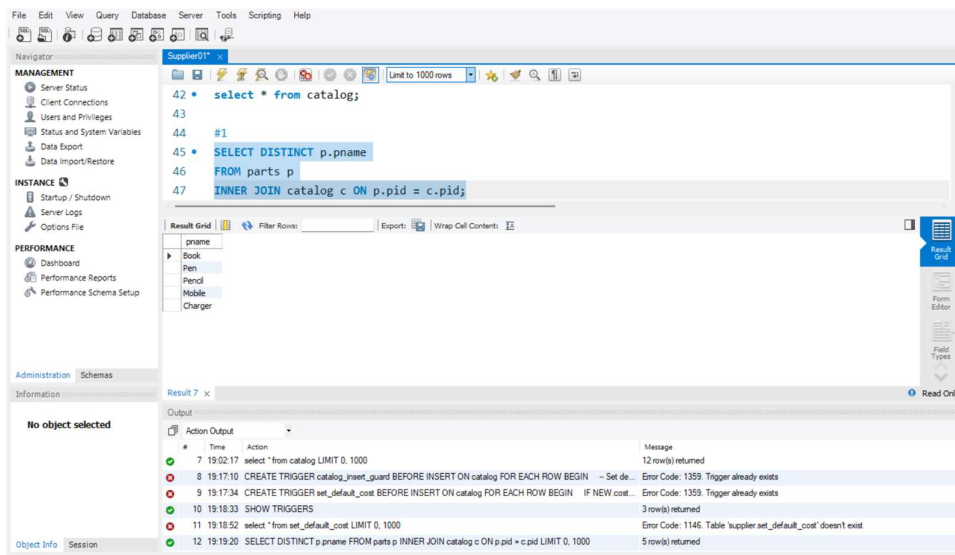
```
insert into catalog values(10001,20001,10);  
insert into catalog values(10001,20002,10);  
insert into catalog values(10001,20003,30);  
insert into catalog values(10001,20004,10);  
insert into catalog values(10001,20005,10);  
insert into catalog values(10002,20001,10);  
insert into catalog values(10002,20002,20);  
insert into catalog values(10003,20003,30);  
insert into catalog values(10004,20003,40);
```

```
select * from catalog;
```

Queries

#1

```
SELECT DISTINCT p.pname  
FROM parts p  
INNER JOIN catalog c ON p.pid = c.pid;
```



#2

SELECT s.sname

FROM supplier s

WHERE NOT EXISTS (

SELECT p.pid

FROM parts p

WHERE NOT EXISTS (

SELECT c.pid

FROM catalog c

WHERE c.sid = s.sid AND c.pid = p.pid));

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

```

54 FROM parts p
55 WHERE NOT EXISTS (
56 SELECT c.pid
57 FROM catalog c
58 WHERE c.sid = s.sid AND c.pid = p.pid))
59

```

Result Grid

Filter Rows: Exports: Wrap Cell Contents: 15

Acme Widget

supplier 8 x

Output

#	Time	Action	Message	Duration / Fetch
8	19:17:10	CREATE TRIGGER catalog_insert_guard BEFORE INSERT ON catalog FOR EACH ROW BEGIN	Error Code: 1359. Trigger already exists	0.000 sec
9	19:17:34	CREATE TRIGGER set_default_cost BEFORE INSERT ON catalog FOR EACH ROW BEGIN IF NEW cost...	Error Code: 1359. Trigger already exists	0.000 sec
10	19:18:33	SHOW TRIGGERS	3 row(s) returned	0.000 sec / 0.000 sec
11	19:18:52	select * from set_default_cost LIMIT 0, 1000	Error Code: 1146. Table 'supplier_set_default_cost' doesn't exist	0.000 sec
12	19:19:20	SELECT DISTINCT p.pname FROM parts p INNER JOIN catalog c ON p.pid = c.pid LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec
13	19:20:19	SELECT s.sname FROM supplier s WHERE NOT EXISTS (SELECT p.pid FROM parts p WHERE NO...	1 row(s) returned	0.000 sec / 0.000 sec

#3

SELECT p.pname

FROM parts p

JOIN catalog c ON p.pid = c.pid

JOIN supplier s ON c.sid = s.sid

GROUP BY p.pname

HAVING COUNT(DISTINCT s.sid) = 1

AND MIN(s.sname) = 'Acme Widget';

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

```

64 JOIN supplier s ON c.sid = s.sid
65 GROUP BY p.pname
66 HAVING COUNT(DISTINCT s.sid) = 1
67 AND MIN(s.sname) = 'Acme Widget';
68
69 #4

```

Result Grid

Filter Rows: Exports: Wrap Cell Contents: 15

Charger
Mobile

Result 9 x

Output

#	Time	Action	Message	Duration / Fetch
9	19:17:34	CREATE TRIGGER set_default_cost BEFORE INSERT ON catalog FOR EACH ROW BEGIN IF NEW cost...	Error Code: 1359. Trigger already exists	0.000 sec
10	19:18:33	SHOW TRIGGERS	3 row(s) returned	0.000 sec / 0.000 sec
11	19:18:52	select * from set_default_cost LIMIT 0, 1000	Error Code: 1146. Table 'supplier_set_default_cost' doesn't exist	0.000 sec
12	19:19:20	SELECT DISTINCT p.pname FROM parts p INNER JOIN catalog c ON p.pid = c.pid LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec
13	19:20:19	SELECT s.sname FROM supplier s WHERE NOT EXISTS (SELECT p.pid FROM parts p WHERE NO...	1 row(s) returned	0.000 sec / 0.000 sec
14	19:21:10	SELECT p.pname FROM parts p JOIN catalog c ON p.pid = c.pid JOIN supplier s ON c.sid = s.sid GROUP BY ...	2 row(s) returned	0.000 sec / 0.000 sec

#4

```
SELECT s.sname
FROM supplier s
WHERE NOT EXISTS (
    SELECT p.pid
    FROM parts p
    WHERE p.color = 'Red'
    AND NOT EXISTS (
        SELECT c.pid
        FROM catalog c
        WHERE c.sid = s.sid AND c.pid = p.pid));
```

The screenshot displays a database management tool interface. The main window shows a SQL query in a text editor, which is the same query as provided in the previous blocks. The query is highlighted in blue. Below the query editor, there is a 'Result Grid' section showing the results of the query. The results are displayed in a table with columns 'sname' and 'pname'. The table contains two rows: 'Acme Widget' and 'Johns'. The interface also includes a sidebar with navigation options like 'MANAGEMENT', 'INSTANCE', and 'PERFORMANCE'. At the bottom, there is an 'Output' section showing the execution log, which includes the query and its results.

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

sname	pname
Acme Widget	
Johns	

supplier 10 x

Output

#	Time	Action	Message	Duration / Fetch
10	19:18:33	SHOW TRIGGERS	3 row(s) returned	0.000 sec / 0.000 sec
11	19:18:52	select * from set_default_cost LIMIT 0, 1000	Error Code 1146: Table 'supplier_set_default_cost' doesn't exist	0.000 sec
12	19:19:20	SELECT DISTINCT p.pname FROM parts p INNER JOIN catalog c ON p.pid = c.pid LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec
13	19:20:19	SELECT s.sname FROM supplier s WHERE NOT EXISTS (SELECT p.pid FROM parts p WHERE NO...	1 row(s) returned	0.000 sec / 0.000 sec
14	19:21:10	SELECT p.pname FROM parts p JOIN catalog c ON p.pid = c.pid JOIN supplier s ON c.sid = s.sid GROUP BY	2 row(s) returned	0.000 sec / 0.000 sec
15	19:21:48	SELECT s.sname FROM supplier s WHERE NOT EXISTS (SELECT p.pid FROM parts p WHERE p.c...	2 row(s) returned	0.016 sec / 0.000 sec

Query Completed

#5

SELECT DISTINCT c.sid

FROM catalog c

WHERE c.cost > (

SELECT AVG(c2.cost)

FROM catalog c2 WHERE c2.pid = c.pid);

The screenshot shows the SQL Server Enterprise Manager interface. The left pane displays the 'Server Enterprise1' tree with nodes for 'MANAGEMENT', 'INSTANCE', and 'PERFORMANCE'. The right pane shows the 'SQL: Enterprise1' window with a query executed. The query is as follows:

```
WHERE c.sid = s.sid AND c.pid = p.pid));
#5
SELECT DISTINCT c.sid
FROM catalog c
WHERE c.cost > (
SELECT AVG(c2.cost)
FROM catalog c2
WHERE c2.pid = c.pid
);
#6
```

The 'Results' grid shows the following data:

sid
10002
10004

The 'Output' pane shows the execution plan and messages. The messages indicate that the query was successful and returned 2 rows.

#	Time	Action	Message	Duration / Fetch
11	19:18:52	select * from set_default_cost LIMIT 0, 1000	Error Code: 1146. Table 'supplier_set_default_cost' doesn't exist	0.000 sec
12	19:19:20	SELECT DISTINCT p.pname FROM parts p INNER JOIN catalog c ON p.pid = c.pid LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec
13	19:20:19	SELECT s.sname FROM suppliers s WHERE NOT EXISTS (SELECT p.pid FROM parts p WHERE NO...	1 row(s) returned	0.000 sec / 0.000 sec
14	19:21:10	SELECT p.pname FROM parts p JOIN catalog c ON p.pid = c.pid JOIN suppliers s ON c.sid = s.sid GROUP BY ...	2 row(s) returned	0.000 sec / 0.000 sec
15	19:21:48	SELECT s.sname FROM suppliers s WHERE NOT EXISTS (SELECT p.pid FROM parts p WHERE p.c...	2 row(s) returned	0.016 sec / 0.000 sec
16	19:22:39	SELECT DISTINCT c.sid FROM catalog c WHERE c.cost > (SELECT AVG(c2.cost) FROM catalog c2 ...	2 row(s) returned	0.000 sec / 0.000 sec

#6

```
SELECT p.pname, s.sname, c.cost
FROM catalog c
JOIN supplier s ON c.sid = s.sid
JOIN parts p ON c.pid = p.pid
WHERE (c.pid, c.cost) IN (
    SELECT pid, MAX(cost)
    FROM catalog
GROUP BY pid);
```

The screenshot shows a SQL IDE interface with a query editor, a result grid, and an output window.

Query Editor: The query is as follows:

```
92 JOIN supplier s ON c.sid = s.sid
93 JOIN parts p ON c.pid = p.pid
94 WHERE (c.pid, c.cost) IN (
95     SELECT pid, MAX(cost)
96     FROM catalog
97     GROUP BY pid);
98
99
100 -- More Queries --
```

Result Grid: The result grid shows the following data:

pname	sname	cost
Book	Acme Widget	10
Book	Johns	10
Book	Acme Widget	10
Book	Acme Widget	10
Pen	Johns	20
Pen	Johns	20
Pencil	Reliance	40
Mobile	Acme Widget	10

Output Window: The output window shows the execution of the query and the results of the subquery. The output is as follows:

#	Time	Action	Message	Duration / Fetch
12	19:19:20	SELECT DISTINCT p.pname FROM parts p INNER JOIN catalog c ON p.pid = c.pid LIMIT 0, 1000	5 row(s) returned	0:00:00 sec / 0:00:00 sec
13	19:20:19	SELECT s.sname FROM supplier s WHERE NOT EXISTS (SELECT p.pid FROM parts p WHERE NO...	1 row(s) returned	0:00:00 sec / 0:00:00 sec
14	19:21:10	SELECT p.pname FROM parts p JOIN catalog c ON p.pid = c.pid JOIN supplier s ON c.sid = s.sid GROUP BY ...	2 row(s) returned	0:00:00 sec / 0:00:00 sec
15	19:21:40	SELECT s.sname FROM supplier s WHERE NOT EXISTS (SELECT p.pid FROM parts p WHERE p.c...	2 row(s) returned	0:01:16 sec / 0:00:00 sec
16	19:22:39	SELECT DISTINCT c.sid FROM catalog c WHERE c.cost > (SELECT AVG(c2.cost) FROM catalog c2 ...	2 row(s) returned	0:00:00 sec / 0:00:00 sec
17	19:23:11	SELECT DISTINCT p.pname, s.sname, c.cost FROM catalog c JOIN supplier s ON c.sid = s.sid JOIN parts p ON c.pid = p...	9 row(s) returned	0:01:15 sec / 0:00:00 sec